Secure Terminal-Based Login System

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Agenda Overview

PROJECT INTRODUCTION

This section provides an overview of the secure terminal-based login system and its significance in enhancing security measures.

TOOLS & TECHNOLOGIES

Here, we discuss the programming language and tools used in developing our secure login system with a focus on security features.

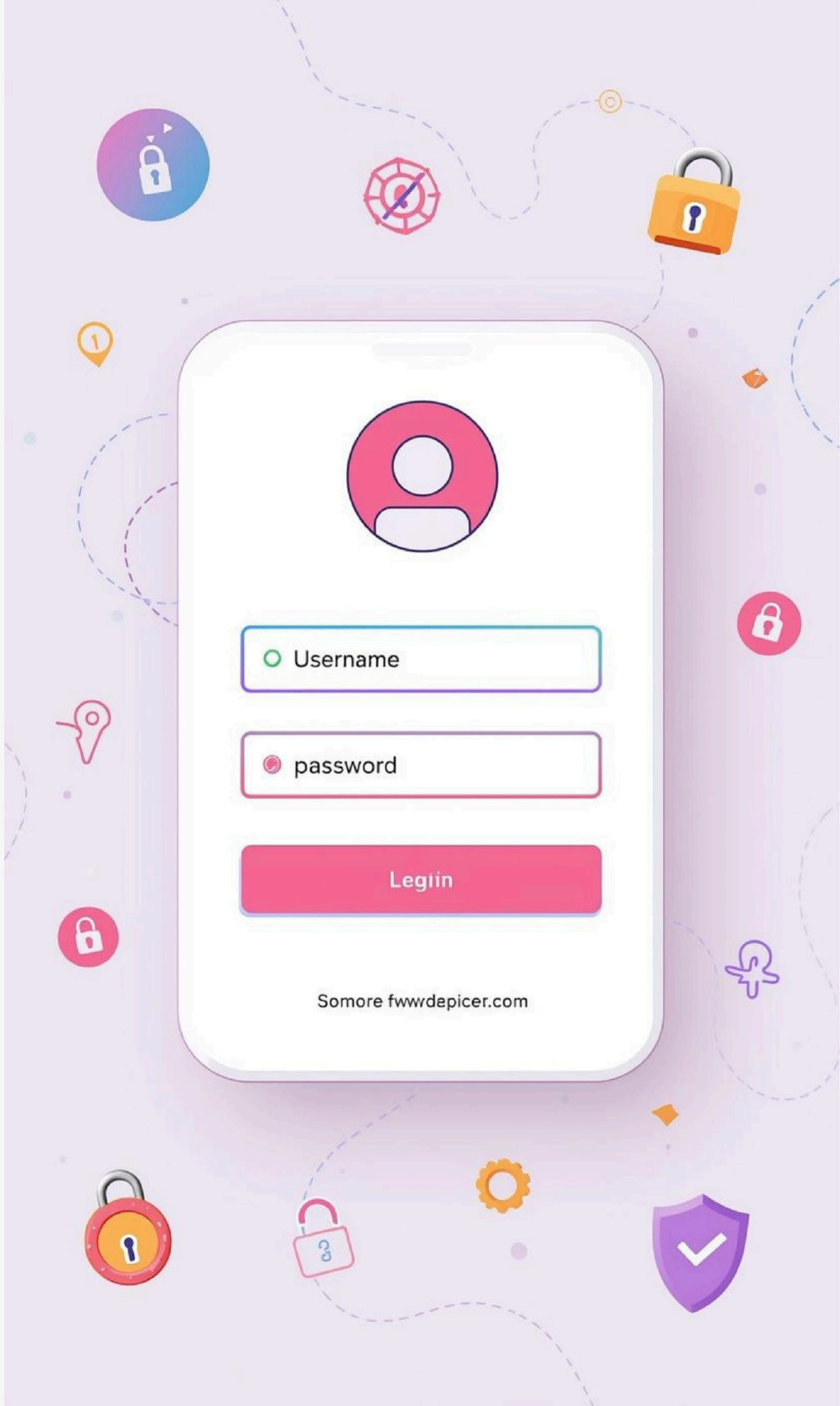
SYSTEM SCREENSHOTS

A detailed walkthrough of the system functionality is provided through screenshots, highlighting user interaction and system responses during the login process.

Purpose of Presentation: Secure Login Systems

IMPORTANCE OF SECURITY

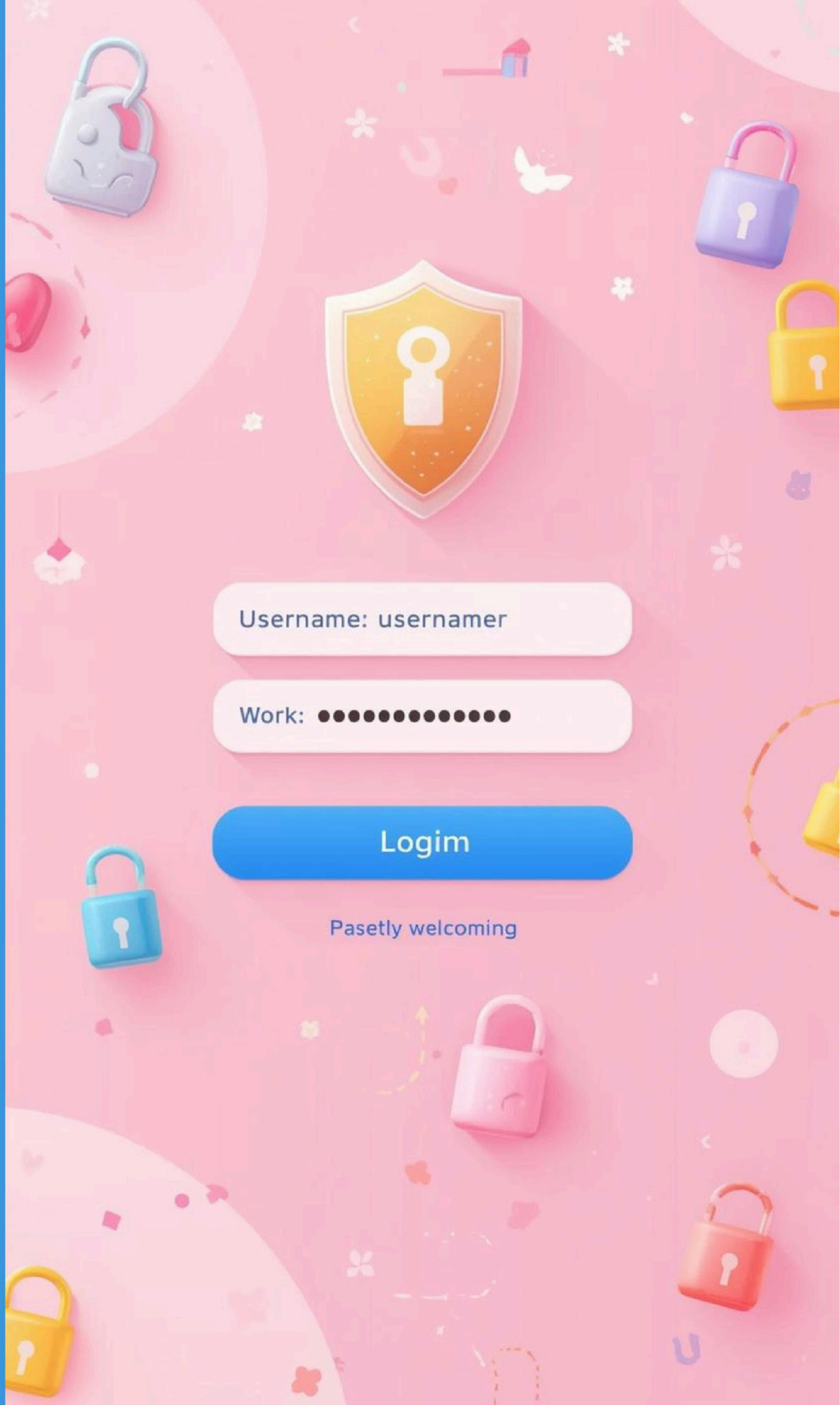
This presentation aims to educate on the significance of secure login systems, including features that enhance **user safety** and prevent unauthorized access.



Introduction to the Secure Terminal-Based Login System

OVERVIEW OF THE SYSTEM

This project focuses on developing a **secure** terminal-based login system that enhances user authentication through multiple security features.



Story of Design Goals

JUNIOR SECURITY ENGINEER'S VISION

The junior security engineer aimed to create a secure terminal-based login system that ensures user safety. The project involves addressing vulnerabilities, implementing robust authentication methods, and providing a user-friendly interface for seamless access while maintaining high-security standards.

Key Security Features

PASSWORD MASKING

Password masking ensures that users' passwords are hidden during entry, enhancing security and preventing unauthorized access to sensitive information.

HASHING

Hashing converts passwords into fixed-length strings, making it nearly impossible to reverse-engineer them, thus protecting user credentials from data breaches.

LIMITED ATTEMPTS

The system limits login attempts to reduce the risk of brute force attacks, automatically locking out users after consecutive failed attempts.

Tools and Technologies Used

PROGRAMMING LANGUAGE

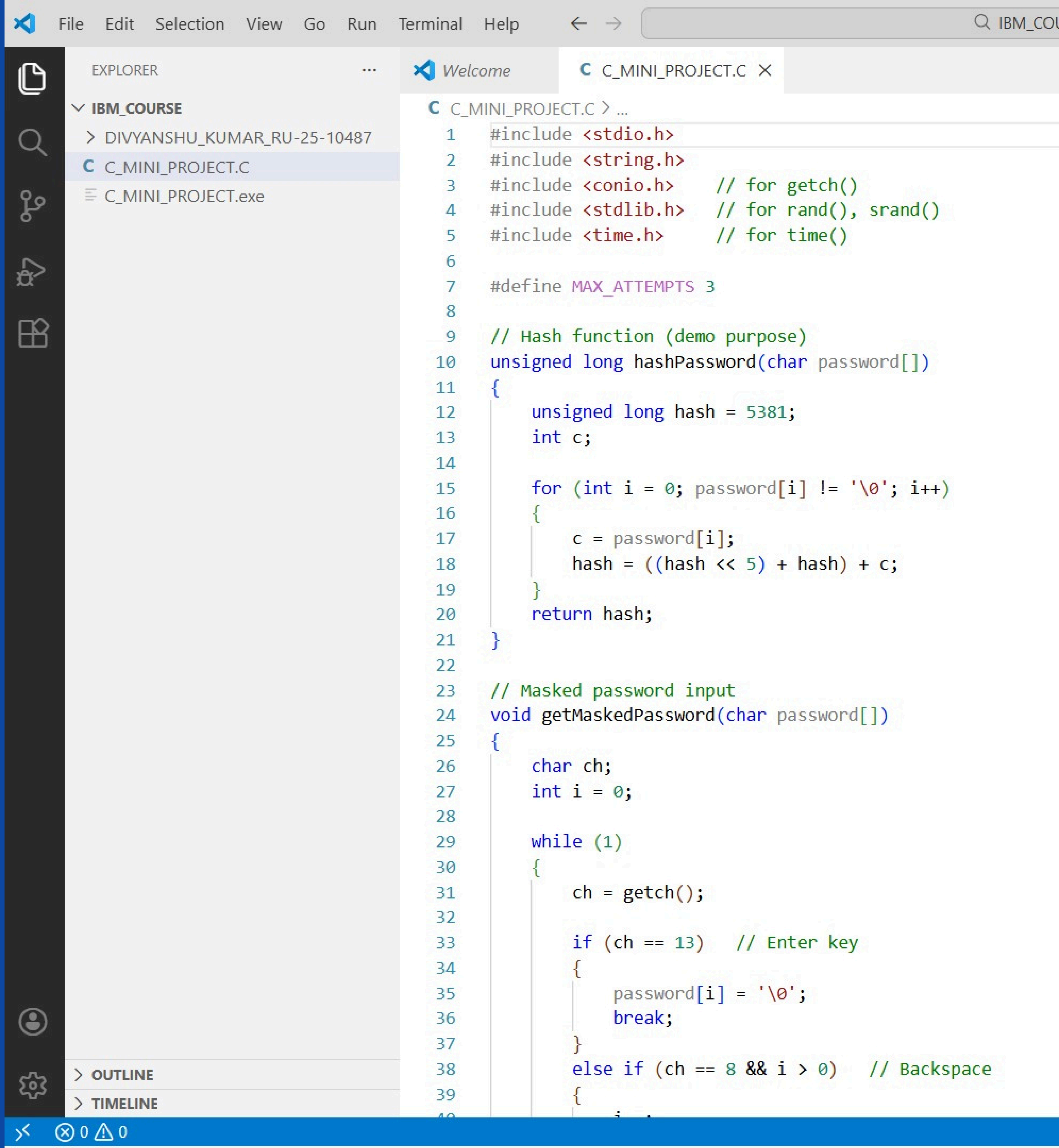
The primary programming language used in this project is **C**, known for its efficiency and control over system resources.

TERMINAL INTERFACE

The project utilizes a terminal interface for user interactions, ensuring a straightforward and efficient way to handle login processes.

Code Page First: Overview and Explanation

- This section includes all the required header files and defines constants for the program.
- A hashing function is implemented to convert passwords into secure hash values.
- Passwords are never stored in plain text, improving system security.
- These functions form the foundation of secure authentication in the project.



Code Page Two Insights

- This function captures the password securely by masking each character with *.
- It uses getch() to read input without displaying actual characters on the screen.
- The function supports backspace handling to allow corrections while typing.
- Password input ends when the Enter key is pressed, ensuring secure entry.

```c
void getMaskedPassword(char password[])
    while (1)
        if (ch == 13)   // Enter key
        else if (ch == 8 && i > 0)   // Backspace
        {
            i--;
            printf("\b \b");
        }
        else
        {
            password[i++] = ch;
            printf("*");
        }
    }
}

int main()
{
    char storedUsername[20] = "Divyanshu10487";
    char registeredMobile[] = "9876543210";
    unsigned long storedPasswordHash = hashPassword("secure10487");

    char username[20], password[20], mobile[15];
    int attempts;
    int loggedIn = 0;   // flag

    while (!loggedIn)
    {
        attempts = 0;

        printf("\n====================================\n");
        printf("   Secure Lab Credential Manager\n");
        printf("====================================\n");

        // LOGIN ATTEMPTS
        while (attempts < MAX_ATTEMPTS)
        {
            printf("Username: ");
            scanf("%s", username);
```

Breakdown of the Third Code Page

- This part of the code handles the main login authentication process.
- It compares the entered username and hashed password with stored credentials.
- The system allows a maximum of three login attempts, displaying remaining attempts each time.
- After three failed attempts, the system gets locked and triggers the forgot password option.

rminal Help IBM_COURSE

Welcome C_MINI_PROJECT.C

C_MINI_PROJECT.C > ...

```c
int main()
    while (!loggedIn)
        while (attempts < MAX_ATTEMPTS)
            printf("Username: ");
            scanf("%s", username);

            printf("Password: ");
            getMaskedPassword(password);
            printf("\n");

            if (strcmp(username, storedUsername) == 0 &&
                hashPassword(password) == storedPasswordHash)
            {
                printf("\n Access Granted!\n");
                printf("Welcome to the secure lab system.\n");
                loggedIn = 1;
                break;
            }
            else
            {
                attempts++;
                printf(" Invalid credentials.\n");
                printf("Attempts left: %d\n\n", MAX_ATTEMPTS - attempts);
            }
        }

        // IF LOCKED
        if (!loggedIn)
        {
            printf(" System Locked!\n");

            char choice;
            printf("Forgot Password? Press Y to continue or N to exit: ");
            scanf(" %c", &choice);

            if (choice == 'Y' || choice == 'y')
            {
                printf("Enter registered mobile number: ");
                scanf("%s", mobile);

```

Code Page Four Analysis

- This code verifies the registered mobile number before allowing password recovery.
- A random OTP is generated using rand() and displayed for verification.
- The user must enter the correct OTP to proceed further.
- Incorrect OTP or unregistered mobile number terminates the reset process.

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Welcome C_MINI_PROJECT.C

C_MINI_PROJECT.C > ...

```c
int main()
    while (!loggedIn)
        if (!loggedIn)
            if (choice == 'Y' || choice == 'y')

                if (strcmp(mobile, registeredMobile) == 0)
                {
                    srand(time(0));
                    int otp = rand() % 9000 + 1000;

                    printf("\n OTP sent successfully!\n");
                    printf("(Demo OTP: %d)\n", otp);

                    int enteredOtp;
                    printf("Enter OTP: ");
                    scanf("%d", &enteredOtp);

                    if (enteredOtp == otp)
                    {
                        printf("\nOTP verified successfully!\n");
                        printf("Enter new password: ");
                        getMaskedPassword(password);
                        printf("\n");

                        storedPasswordHash = hashPassword(password);
                        printf(" Password reset successful!\n");
                        printf("Please login again.\n");
                    }
                    else
                    {
                        printf(" Incorrect OTP.\n");
                        break;
                    }
                }
                else
                {
                    printf(" Mobile number not registered.\n");
                    break;
                }
```

Final Code Page Overview

- This part of the code handles the Forgot Password logic after the system is locked.
- It verifies the registered mobile number and generates a One-Time Password (OTP).
- Only when the correct OTP is entered, the user is allowed to reset the password securely.
- After password reset, the system redirects the user to login again for authentication.

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Welcome | C_MINI_PROJECT.C

C_MINI_PROJECT.C > ...

```c
int main()
    while (!loggedIn)
        if (!loggedIn)
            if (choice == 'Y' || choice == 'y')
                if (strcmp(mobile, registeredMobile) == 0)

                        storedPasswordHash = hashPassword(password);
                        printf(" Password reset successful!\n");
                        printf("Please login again.\n");
                    }
                    else
                    {
                        printf(" Incorrect OTP.\n");
                        break;
                    }
                }
                else
                {
                    printf(" Mobile number not registered.\n");
                    break;
                }
            }
            else
            {
                printf("System exiting...\n");
                break;
            }
        }
    }

    return 0;
}
```

Secure Login Screen with Password Masking

ENHANCED USER SECURITY

This login screen features **password masking** to protect sensitive information, ensuring users can securely enter their credentials without revealing them.

```
====================================
   Secure Lab Credential Manager
====================================
Username: Divyanshu10487
Password: ***********
```

Successful Login: Access Granted

USER EXPERIENCE ACHIEVED

Upon entering the correct credentials, users receive a confirmation of successful login, granting them access to the secure terminal features.

```
====================================
   Secure Lab Credential Manager
====================================
Username: Divyanshu10487
Password: ***********

 Access Granted!
Welcome to the secure lab system.
PS C:\Users\divya\OneDrive\Documents\IBM_COURSE>
```

Failed Login Attempts Resulting in Lockout

SECURITY MEASURES IN ACTION

After three failed login attempts, the system automatically locks the user out, demonstrating **robust security** against unauthorized access and potential attacks.

```
====================================
   Secure Lab Credential Manager
====================================
Username:  divyanshu10487
Password: ***********
 Invalid credentials.
Attempts left: 2

Username: Divyanshu10487
Password: ***********
 Invalid credentials.
Attempts left: 1

Username:  divyanshu10487
Password: ***********
 Invalid credentials.
Attempts left: 0
```

Locked System Prompt: Forgot Password Option

USER ASSISTANCE AND RECOVERY

The system prompts users with a **clear message** about their account being locked after multiple failed attempts, offering a secure password recovery option.

```
========================================
   Secure Lab Credential Manager
========================================
Username:  divyanshu10487
Password: ***********
 Invalid credentials.
Attempts left: 2

Username: Divyanshu10487
Password: ***********
 Invalid credentials.
Attempts left: 1

Username:  divyanshu10487
Password: ***********
 Invalid credentials.
Attempts left: 0

 System Locked!
```

OTP Generation and Input Simulation

SECURE ONE-TIME PASSWORD PROCESS

The OTP generation process ensures enhanced security, allowing users to receive a temporary code for authentication during the login process.

```
====================================
   Secure Lab Credential Manager
====================================
Username: Divyanshu10487
Password: ***********
 Invalid credentials.
Attempts left: 2

Username:  divyanshu10487
Password: ******
 Invalid credentials.
Attempts left: 1

Username:  divyanshu10487
Password: *********
 Invalid credentials.
Attempts left: 0

 System Locked!
Forgot Password? Press Y to continue or N to exit: Y
Enter registered mobile number: 9876543210

 OTP sent successfully!
(Demo OTP: 8743)
Enter OTP: 8743
```

Password Reset Confirmation Process

SUCCESSFUL PASSWORD UPDATE

After entering the new password, users receive a confirmation message, ensuring security and ease of access to their accounts moving forward.

```
====================================
   Secure Lab Credential Manager
====================================
Username:  divyanshu10487
Password: ***********
 Invalid credentials.
Attempts left: 2

Username: Divyanshu10487
Password: ***********
 Invalid credentials.
Attempts left: 1

Username:  divyanshu10487
Password: ***********
 Invalid credentials.
Attempts left: 0

 System Locked!
Forgot Password? Press Y to continue or N to exit:
Enter registered mobile number: 9876543210
```

Redirecting to the Login Screen After Reset

EASY ACCESS RESTORED

After a successful password reset, users are redirected to the login screen, ensuring a **smooth transition** and convenient access to the system.

```
====================================
   Secure Lab Credential Manager
====================================
Username: Divyanshu10487
Password: ***********
 Invalid credentials.
Attempts left: 2

Username:  divyanshu10487
Password: ******
 Invalid credentials.
Attempts left: 1

Username:  divyanshu10487
Password: *********
 Invalid credentials.
Attempts left: 0

 System Locked!
Forgot Password? Press Y to continue or N to exit: Y
Enter registered mobile number: 9876543210

 OTP sent successfully!
(Demo OTP: 8743)
Enter OTP: 8743

OTP verified successfully!
Enter new password: *********
 Password reset successful!
Please login again.

====================================
   Secure Lab Credential Manager
====================================
Username: Divyanshu10487
Password: *********

 Access Granted!
Welcome to the secure lab system.
PS C:\Users\divya\OneDrive\Documents\IBM_COURSE>
```

Real-World Applications

SECURE LAB ENVIRONMENTS

The login system ensures **robust security** in research labs, protecting sensitive data from unauthorized access and potential breaches.

RESEARCH DATA PROTECTION

It plays a crucial role in safeguarding **research data**, providing a reliable way to manage user access and maintain confidentiality.

SMALL-SCALE SYSTEMS

This system is ideal for **small-scale setups**, enabling organizations to implement effective security measures with minimal complexity and cost.

Advantages of the System

SECURITY

Enhanced security is achieved through **password hashing**, making it difficult for attackers to retrieve sensitive login information.

USABILITY

User-friendly features, such as **password masking** and OTP resets, improve the overall experience for users while enhancing security.

PROTECTION

Attack mitigation strategies, including **limited login attempts** and account lockouts, protect the system from unauthorized access and brute force attacks.

Future Enhancements

BIOMETRIC INTEGRATION

Implementing biometric authentication can enhance security, enabling users to log in using fingerprints or facial recognition for seamless access.

SMS OTP

Real-time SMS OTP sending can verify user identities, adding an extra layer of security during the login process for sensitive systems.

GUI INTERFACE

Developing a graphical user interface would improve user experience, making the system more intuitive for users unfamiliar with command-line interfaces.

Future Scalability

LARGER SYSTEMS

The secure terminal-based login system can be expanded to accommodate larger user bases and more complex organizational structures effectively.

ENTERPRISE USE

This system is suitable for enterprise environments, providing robust security features essential for protecting sensitive corporate data and user information.

LOGGING AND AUDIT

Implementing logging and audit trails enhances security by tracking user activities, helping identify potential breaches and ensuring compliance with regulations.

Recap of Presentation

PURPOSE

The login system aims to provide **secure access** to sensitive data, ensuring user authentication and safeguarding against unauthorized entry.

KEY FEATURES

Major features include **password masking**, account lockout after failed attempts, and OTP-based reset to enhance security and user experience.

BENEFITS

This system promotes **data protection**, user-friendly interfaces, and mitigates unauthorized access risks, making it essential for secure application environments.

Thank You for Your Attention

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